FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

Urinary Tract Infections

Tutorial | Week 6 · Tetanus, skin, poisons & hormones | Paediatrics | Dr Nabatte

UTI is a common hidden cause of fever in infants and young children; the diagnosis rests on a properly collected urine sample (clean catch, catheter or suprapubic aspiration, not a bag culture) with dipstick or microscopy and culture, babies under 3 months and sick children get parenteral antibiotics, older children get oral treatment for 3 days (cystitis) or 7–10 days (pyelonephritis), and imaging is targeted at infants and at atypical or recurrent infections to find obstruction and vesicoureteric reflux.

Must know before the ward round

1. Test the urine of every febrile infant or young child with no obvious source (a negative malaria test does not end the search). Half of infants with UTI have fever and nothing else.
2. Collection: clean catch is first choice; urine pad if not possible (never cotton wool); catheter or suprapubic aspiration (SPA) if non-invasive methods fail or the child is very sick. A bag sample is only useful if it is negative.
3. Culture thresholds: SPA any growth of a uropathogen; catheter ≥10,000–50,000 CFU/mL (AAP: ≥50,000); clean catch ≥100,000 CFU/mL of a single organism. Culture within 4 hours, or refrigerate.
4. Dipstick (NICE NG224): from 3 months, leucocyte esterase or nitrite positive → send culture and treat; both negative → UTI unlikely. Under 3 months: urgent microscopy and culture, not dipstick alone. Nitrite is specific but often negative in infants (they void too often).
5. Upper UTI (pyelonephritis) = bacteriuria with fever ≥38 °C, or loin pain/tenderness. Bacteriuria without systemic signs = lower UTI (cystitis).
6. Under 3 months: admit and give IV antibiotics (ampicillin plus gentamicin in neonates; e.g. ceftriaxone or cefotaxime plus ampicillin at 1–3 months), with blood culture and a low threshold for LP.
7. Over 3 months, pyelonephritis: oral cefalexin, co-amoxiclav or cefixime for 7–10 days if well (UCG 2023: 10 days); IV ceftriaxone 50–80 mg/kg once daily or gentamicin 7.5 mg/kg once daily if vomiting or seriously ill, then oral to complete.
8. Over 3 months, cystitis: oral nitrofurantoin or trimethoprim/cefalexin for 3 days (NICE); UCG 2023: nitrofurantoin 3 mg/kg/day in 4 doses for 7 days. Amoxicillin and co-trimoxazole resistance is high in Ugandan E. coli.
9. Atypical UTI (NICE): seriously ill, poor urine flow, abdominal or bladder mass, raised creatinine, septicaemia, no response within 48 h, non-E. coli organism. Recurrent UTI: ≥2 upper, or 1 upper + ≥1 lower, or ≥3 lower.
10. Imaging (NICE): under 6 months, ultrasound within 6 weeks even if typical; atypical UTI, ultrasound during the illness; DMSA 4–6 months later for atypical/recurrent UTI under 3 years and recurrent UTI at any age; MCUG for atypical/recurrent UTI under 6 months.
11. VUR grades I–V: low grades (I–II) usually resolve with growth; high grades (IV–V) and bilateral reflux carry the most risk of scarring, hypertension and CKD. RIVUR (2014): prophylaxis halved recurrences but did not reduce scarring.
12. An infant boy with UTI, poor stream, a palpable bladder or bilateral hydronephrosis has posterior urethral valves until proven otherwise: catheterise and refer for MCUG and paediatric urology.

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In a nutshell

A **urinary tract infection (UTI)** is bacterial infection anywhere from the urethra (urethritis) and bladder (**cystitis**, lower UTI) to the kidney (**pyelonephritis**, upper UTI). In older children it looks like an adult UTI: dysuria, frequency and loin pain. In infants it hides as **fever without a focus**, vomiting, poor feeding, irritability or jaundice.

It matters because:

1. **It is common and missed.** In Uganda every fever is first a malaria question. A febrile child with a negative RDT and no focus needs a urine test.
2. **It can kill** young infants through **urosepsis**.
3. **It can scar the kidney.** Repeated pyelonephritis, especially with **vesicoureteric reflux (VUR)** or **obstruction** (posterior urethral valves), leads to scarring, hypertension and chronic kidney disease.

> **CLINICAL PEARL**
> The one thing never to forget: **the diagnosis is only as good as the urine sample.** A contaminated bag sample leads to wrong antibiotics and unnecessary scans. Take a clean catch (or catheter/SPA in sick infants) **before** the first antibiotic dose whenever you can, but never let sample collection delay treatment of a septic infant.

Definitions & classification

Term	Definition
UTI	Symptoms or signs **plus** significant growth of a uropathogen from a properly collected urine sample (with pyuria in most)
Upper UTI (acute pyelonephritis)	Bacteriuria **and fever ≥38 °C**, or bacteriuria with fever <38 °C **and loin pain/tenderness** (NICE NG224)
Lower UTI (cystitis)	Bacteriuria without systemic features (dysuria, frequency, suprapubic pain)
Atypical UTI (NICE)	**Seriously ill**; **poor urine flow**; **abdominal or bladder mass**; **raised creatinine**; **septicaemia**; **failure to respond within 48 h**; **non-*E. coli* organism**
Recurrent UTI (NICE)	**≥2 episodes of upper UTI**, or **1 upper + ≥1 lower**, or **≥3 lower UTIs**
Asymptomatic bacteriuria	Bacteriuria without symptoms; **do not treat** (except in specific settings such as pregnancy or before urological surgery)
Vesicoureteric reflux (VUR)	Retrograde flow of urine from the bladder into the ureter and kidney; **primary** (short intravesical ureter) or **secondary** (high bladder pressure: posterior urethral valves, neurogenic bladder, bladder–bowel dysfunction)
Bladder and bowel dysfunction (BBD)	Voiding dysfunction (holding, infrequent voiding, incomplete emptying, wetting) with **constipation**; a major cause of recurrent UTI in school-age children

Significant culture growth by collection method

Method	Significant growth	Comment
Suprapubic aspiration (SPA)	**Any growth** of a Gram-negative bacillus (or more than a few thousand CFU/mL of Gram-positive cocci)	Gold standard; ultrasound-guided
Urethral catheter	**≥10,000 CFU/mL** (AAP 2011 uses **≥50,000 CFU/mL** for febrile infants 2–24 months)	Low contamination if done aseptically
Clean catch / midstream	**≥100,000 CFU/mL** of a single organism	10,000–100,000 in a symptomatic child = repeat the sample
Urine bag	Positive culture is **unreliable** (false-positive rate very high); **negative culture is useful**	Use only to screen; confirm a positive with a clean catch or catheter
Mixed growth (≥2 organisms)	Usually contamination	Repeat sample

Epidemiology & Uganda context

- **How common?** About **5–7% of febrile children aged 2–24 months** with no obvious source have a UTI (AAP). By age 7, roughly **8% of girls and 2% of boys** have had one.
- **Sex and age**: in the first **3–6 months**, **boys** (especially uncircumcised) are affected more often, and **posterior urethral valves** present here. After infancy, **girls** predominate (short urethra).
- **Organisms**: ***E. coli* (80–90%)**; *Klebsiella*, *Proteus* (boys, alkaline urine, stones), *Enterococcus* (infants, abnormal tracts), *Pseudomonas* (catheters, abnormal tracts), *Staphylococcus saprophyticus* (adolescent girls). **ESBL-producing *E. coli* and *Klebsiella*** are increasingly common.
- **Risk factors**: infancy, female sex, **uncircumcised boys** (first year), **constipation**, **voiding dysfunction**, **VUR**, **obstruction** (PUV, pelviureteric junction obstruction), **neurogenic bladder** (spina bifida), **catheters**, **severe acute malnutrition**, **HIV**, sexual activity (adolescents), **sexual abuse**, **schistosomiasis** (bladder damage).

UGANDA CONTEXT

The Ugandan reality. Urine dipsticks are available at most HC IIIs and above; urine microscopy at HC IV level; **culture and sensitivity** mostly at **RRHs, Mulago, Mengo, Lacor** and private labs (results take 2–3 days). Ugandan hospital studies report **high *E. coli* resistance to amoxicillin/ampicillin and co-trimoxazole**, and a substantial share of **ESBL producers**, so **empirical amoxicillin or co-trimoxazole is a poor choice** for pyelonephritis. **Ultrasound** is available at most hospitals and many HC IVs; **MCUG** at Mulago and some private/mission hospitals; **DMSA** only at Mulago nuclear medicine (limited). Children with valves or high-grade VUR need **paediatric urology/surgery and nephrology** (Mulago and other national centres).

Pathophysiology

1. **Ascending infection** (the usual route after the newborn period): bowel flora colonise the **periurethral area** → ascend the urethra into the bladder. Girls' short urethra, uncircumcised boys' foreskin (a reservoir in infancy), poor hygiene and wiping back-to-front all help.
2. **Bladder defences**: regular **complete emptying** washes bacteria out. **Infrequent voiding, incomplete emptying and constipation** (a full rectum compresses the bladder neck and causes residual urine) undermine this, which is why **BBD** causes recurrent UTI.
3. **Ascent to the kidney**: *E. coli* with **P-fimbriae** bind to uroepithelium and reach the kidney, especially with **VUR**, which carries infected urine to the renal papillae ("intrarenal reflux" at the poles).

4. **Renal damage**: inflammation in the parenchyma → **scarring** (seen on DMSA) → loss of nephrons → **hypertension, proteinuria and CKD** (reflux nephropathy). The **younger the child**, the more vulnerable the kidney, and the **longer the delay** in treatment, the more scarring.
5. **Haematogenous spread** (neonates): bacteraemia seeds the kidney, so **neonatal UTI is often part of sepsis**, and meningitis may coexist.
6. **Obstruction** (PUV, PUJ obstruction, stones) creates stagnant, infected urine and high pressure → severe infection, hydronephrosis and renal impairment.

VUR grading (International Reflux Study, on MCUG)

Grade	Findings	Natural history
I	Reflux into the **ureter only**	Most resolve
II	Into ureter, **pelvis and calyces, no dilatation**	Most resolve (about 80% of grades I–II)
III	**Mild–moderate dilatation** of ureter and pelvis, slight blunting of fornices	About half resolve
IV	**Moderate dilatation and tortuosity** of the ureter, obliteration of fornices, papillary impressions preserved	Less likely to resolve; scarring risk
V	**Gross dilatation and tortuosity**, loss of papillary impressions	Rarely resolve; highest risk of scarring and CKD; often surgical

Clinical acumen

Focused history

Symptoms by age (NICE NG224)

Age	Most common	Less common
Under 3 months	**Fever**, vomiting, lethargy, irritability	Poor feeding, failure to thrive, abdominal pain, **jaundice**, haematuria, offensive urine
3 months and over, preverbal	**Fever**	Abdominal pain, loin tenderness, vomiting, poor feeding, lethargy, irritability, haematuria, offensive urine, failure to thrive
Verbal child	**Frequency, dysuria**	Dysfunctional voiding, **new wetting** (day or night), abdominal or loin pain, fever, malaise, vomiting, haematuria, offensive or cloudy urine

Question	Why it matters
Fever: duration, height; malaria test done?	Fever without a source → urine test
Vomiting, feeding, lethargy, jaundice (infant)	Sepsis risk; IV treatment
Dysuria, frequency, urgency, new wetting, smelly urine	Lower UTI
Loin or abdominal pain	Upper UTI
Urine stream in a boy (weak, dribbling, straining)	**Posterior urethral valves**
Previous UTIs, unexplained fevers	Recurrent UTI → imaging, prophylaxis
Antenatal scan abnormal (hydronephrosis)	Congenital anomaly
Bowel habit: constipation, hard stools, soiling	BBD
Voiding habits: holding, infrequent voiding, school toilet avoidance	BBD

Question	Why it matters
Family history of VUR or kidney disease	VUR is familial (siblings)
Growth, BP history	Renal scarring, CKD
HIV, SAM, sickle cell disease, spina bifida, catheter	Higher risk, atypical organisms
In girls: vaginal discharge, genital itch; in adolescents: sexual activity	Vulvovaginitis, STI, **sexual abuse** (ask sensitively and safeguard)
Water contact (lake/river), terminal haematuria	Schistosomiasis

Examination

1. **ETAT**: airway, breathing, **circulation (sepsis/shock)**, consciousness, glucose in a sick infant.
2. **Vital signs for age**, **temperature**, **blood pressure** (with the right cuff size: hypertension suggests scarring or renal disease).
3. **Growth**: weight, length/height, Z-scores, MUAC (failure to thrive, SAM).
4. **Abdomen**: **loin tenderness**, **palpable kidney** (hydronephrosis), **palpable bladder** (outflow obstruction), constipation (faecal masses), other masses (Wilms tumour).
5. **Genitalia**: phimosis, meatal stenosis, labial adhesions, vulvovaginitis, signs of abuse; observe the **urine stream** in boys if possible.
6. **Spine and lower limbs**: sacral dimple, hair tuft, lipoma, neurological deficit (**neurogenic bladder**).
7. **Look for another source of fever**: ears, throat, chest, meninges, skin.

Red flags

RED FLAG

- **Infant under 3 months** with fever (sepsis risk; refer and treat IV)
- **Septic or shocked** child, vomiting everything, lethargy
- **Poor urine stream, palpable bladder** or bilateral loin masses in a boy (posterior urethral valves)
- **Raised creatinine**, oliguria, hypertension
- **No response after 48 h** of appropriate antibiotics (abscess, obstruction, resistant organism)
- **Recurrent UTI**, poor growth
- In a girl with recurrent dysuria and genital signs: **possible sexual abuse** (follow child protection procedures)

Thinking like a clinician

- **Fever with a negative RDT and no focus in a child under 2 → urine**, before blaming "viral fever".
- **Infant boy with UTI** → look at the **stream**, feel for a **bladder**, scan the kidneys: **PUV** must be excluded.
- **School-age girl with recurrent cystitis and wetting** → ask about **constipation and holding**: treat **BBD**, not just the infection.
- **Dipstick positive for blood only**, no leucocytes or nitrites → consider **haematuria** causes (schistosomiasis, glomerulonephritis, stones, trauma), not UTI.
- **Positive bag culture with mixed growth → contamination**, not UTI; recollect properly.
- **"Recurrent UTI" on paper** → check how each diagnosis was made; many are contaminated samples.
- **Common student mistakes**: using a bag culture to diagnose; using CRP alone to decide upper vs lower UTI (NICE says don't); stopping antibiotics after 3 days for pyelonephritis; prescribing amoxicillin in a high-resistance setting; giving nitrofurantoin for pyelonephritis (it does not reach the kidney); forgetting the BP; ordering an MCUG for every first UTI; forgetting to treat constipation.

EXAM FAVOURITE

Viva favourite: **"How would you collect urine from a 9-month-old?"** Answer: **clean catch** first (sit with the nappy off, clean the genitalia with water, wait with a sterile pot; stimulation techniques help). If not possible, a **urine collection pad** (not cotton wool or gauze). If the child is very sick or non-invasive methods fail, **catheter sample or ultrasound-guided SPA**. A **bag** sample is acceptable only to **rule out** UTI when negative.

Clinical workup

Step 0: First 5 minutes

1. **ETAT**: airway, breathing, **circulation** (cold hands, CRT >3 s, weak fast pulse = shock), coma, convulsions; **glucose** in a sick infant.
2. **Septic or under 3 months?** → IV/IO access, **blood culture**, glucose, FBC, creatinine; collect **urine by catheter or SPA** quickly; **first dose of IV antibiotics without waiting** for the sample if it will delay treatment; consider **LP** (especially neonates).
3. **Shock** → 10–20 mL/kg crystalloid and reassess (not in SAM; neonates 10 mL/kg).
4. **Well child** → collect a clean urine sample **before** antibiotics and test it.

Step 1: Focused history

- [ ] **Fever pattern and malaria test** → fever without a focus.
- [ ] **Urinary symptoms, wetting, smell** → lower vs upper UTI.
- [ ] **Stream (boys)**, previous UTIs, antenatal scans → structural abnormality.
- [ ] **Constipation, voiding habits** → BBD.
- [ ] **Family history** of VUR, kidney disease → screening and imaging.
- [ ] **HIV, SAM, SCD, neurological conditions, catheter** → atypical organisms, severity.
- [ ] **Immunisation, feeding, growth** → overall child health.
- [ ] **Sexual history (adolescent) / safeguarding (younger)** → STI or abuse.

Step 2: Focused examination

- **General**: temperature, HR, RR, **BP**, CRT, hydration, jaundice, weight, length, Z-scores, MUAC.
- **Abdomen**: loin tenderness, renal masses, bladder, faecal loading.
- **Genitalia and urine stream**; **spine**; **lower-limb neurology**.
- **Other foci**: ears, throat, chest, meninges, skin.
- **Document negatives**: "no palpable bladder or kidneys, normal stream, spine normal, BP 92/58 (50th centile for age)".

Step 3: Bedside tests

Test	How	What it tells you
Urine dipstick (fresh sample)	Read at the times printed on the bottle	**Leucocyte esterase** (pyuria; sensitive ~80%, less specific) and **nitrite** (specific >90% but **sensitivity ~50%**; needs about 4 h in the bladder and nitrate-reducing organisms, so often negative in infants and with *Enterococcus*/ *Pseudomonas*); blood and protein (non-specific)
Urine microscopy (fresh, uncentrifuged)	Count WBC and look for bacteria	**Pyuria ≥10 WBC/mm³** (WHO Pocket Book: >5 WBC per high-power field) and **bacteriuria**; both positive = UTI

Test	How	What it tells you
Gram stain of urine (if lab available)		Gram-negative rods (*E. coli*) vs Gram-positive cocci (*Enterococcus*, *Staph*)
Blood glucose	Sick infant	Hypoglycaemia
Malaria RDT		Co-existing malaria
Bladder scan / bedside ultrasound (if skilled)		Full bladder (for SPA), residual urine, hydronephrosis

Dipstick interpretation (NICE NG224)

Result	Children 3 months to 3 years	Children 3 years and over
LE + and nitrite +	Treat as UTI; **send culture**	Treat as UTI; send culture if high/intermediate risk of serious illness or previous UTI
LE – and nitrite +	Treat; send culture	**Treat** (if fresh sample); send culture
LE + and nitrite –	Treat; send culture	**Send culture**; treat only if there is good clinical evidence of UTI
LE – and nitrite –	UTI unlikely: no antibiotics, no culture (unless other criteria)	UTI unlikely: no antibiotics; look for another diagnosis

Under 3 months: do **urgent microscopy and culture**; do not rely on a dipstick.

Step 4: Laboratory & imaging

Test	When	Expected / abnormal finding	How it changes management
Urine culture and sensitivity	Under 3 months; suspected upper UTI; high/intermediate risk of serious illness; dipstick and clinical picture disagree; recurrent UTI; no response	**Single organism** above the threshold for the collection method; sensitivities	Confirms diagnosis; adjusts antibiotics (ESBL); defines recurrence. RRH/NRH, Mengo, Lacor, private labs; refrigerate if delayed (process within 4 h, or within 24 h refrigerated)
Blood culture	Infants, septic children	Same organism (urosepsis)	Duration and route of antibiotics
FBC, CRP	Sick child	Leucocytosis; raised CRP (cannot alone separate upper from lower UTI)	Severity, response
Urea, creatinine, electrolytes	Infants, upper UTI, atypical UTI, abnormal imaging	Raised creatinine (obstruction, dehydration, CKD); hyponatraemia/hyperkalaemia (obstructive uropathy)	Atypical UTI → imaging; gentamicin dosing; nephrology
LP	Febrile neonate; infant under 3 months who is unwell	Meningitis co-existing	Antibiotic choice and duration
Renal and bladder ultrasound (USS)	Per NICE tables (below); any infant boy with UTI; atypical/recurrent UTI	**Hydronephrosis**, **dilated ureters**, **thick-walled bladder** (PUV, neurogenic), small or scarred kidney, abscess, stones, duplex kidney	Detects obstruction needing urgent drainage; guides MCUG. Most hospitals and many HC IVs

Test	When	Expected / abnormal finding	How it changes management
MCUG / VCUG (micturating cystourethrogram)	Atypical or recurrent UTI under 6 months; suspected PUV; dilatation on USS	**VUR grade**; **dilated posterior urethra with valves**	Prophylaxis, surgery. Mulago, some private/mission hospitals. Give 3 days of prophylactic antibiotics with the MCUG on day 2 (NICE)
DMSA scan	4–6 months after atypical/recurrent UTI (per age, NICE)	**Renal scars**, differential function	Long-term follow-up (BP, proteinuria). Mulago nuclear medicine only
Urine for schistosome eggs	Haematuria with water contact	Terminal-spine eggs	Praziquantel
HIV test	Unknown status, recurrent infections		ART, cotrimoxazole

NICE NG224 imaging schedule

Test	Under 6 months (good response / atypical / recurrent)	6 months–3 years	3 years and over
USS during acute infection	No / **Yes** / **Yes**	No / **Yes** / No	No / **Yes** / No
USS within 6 weeks	**Yes** (if abnormal consider MCUG) / No / No	No / No / **Yes**	No / No / **Yes**
DMSA 4–6 months later	No / **Yes** / **Yes**	No / **Yes** / **Yes**	No / No / **Yes**
MCUG	No / **Yes** / **Yes**	Not routinely (consider if dilatation on USS, poor urine flow, non-*E. coli* infection, family history of VUR)	No

AAP 2011 (febrile UTI, 2–24 months; reaffirmed): **renal and bladder ultrasound after the first febrile UTI**; **VCUG only if the ultrasound shows hydronephrosis, scarring or other findings suggesting high-grade VUR or obstruction**, or after a **second** febrile UTI.

Step 5: Putting it together

Model one-line summary

> "A 4-month-old boy, 6.1 kg (weight-for-age Z −0.9), fully immunised for age, exclusively breastfed, HIV-unexposed, presenting with 2 days of fever 39.6 °C, vomiting and irritability, RDT negative, with a catheter urine showing leucocytes 3+, nitrite positive and >50 WBC/mm³ with Gram-negative rods, and a palpable bladder: febrile UTI (presumed pyelonephritis) in a young infant with possible bladder outflow obstruction."

Problem list: (1) febrile UTI under 6 months (IV antibiotics); (2) palpable bladder in a boy → exclude **posterior urethral valves**; (3) creatinine to check; (4) imaging plan (USS now; MCUG).

EXAM FAVOURITE

Worked example 1: interpreting a culture.

- **Bag urine** from a 10-month-old: "mixed growth of 3 organisms, 10^5 CFU/mL" → **contamination**; recollect by clean catch or catheter.
- **Catheter urine**: "*E. coli* 60,000 CFU/mL, pyuria present" → **UTI** (above the AAP threshold of 50,000 for catheter samples).
- **Clean catch** in a 6-year-old with dysuria: "*E. coli* 20,000 CFU/mL" → borderline; repeat the sample; treat if symptomatic with pyuria.
- **SPA**: "*Klebsiella* 2,000 CFU/mL" → **any growth from SPA is significant**; also a **non-*E. coli*** organism, so this is an **atypical UTI** (imaging needed).

EXAM FAVOURITE

Worked example 2: dose calculation. A 2-year-old girl, 12 kg, with pyelonephritis is vomiting: **ceftriaxone 50–80 mg/kg once daily** (UCG) → **600–960 mg IV**; most units give **75 mg/kg ≈ 900 mg**. After 48 h she is eating and afebrile: switch to **cefixime 8 mg/kg once daily = 96 mg (≈100 mg)** to complete **10 days** in total. If you use gentamicin instead: **7.5 mg/kg once daily = 90 mg IV/IM**, checking creatinine (avoid in renal impairment).

Differential diagnosis

Condition	Distinguishing features	Key investigation
Malaria	Fever, positive RDT; may coexist	RDT/slide; urine still tested if no focus
Meningitis / sepsis	Infant with fever, irritability, bulging fontanelle	LP, blood culture
Pneumonia	Cough, fast breathing, indrawing	Clinical, SpO_2
Gastroenteritis	Diarrhoea with vomiting; pyuria can occur with dehydration	Urine culture if fever persists
Appendicitis	Migrating right iliac fossa pain, guarding; mild pyuria possible	Clinical, USS
Vulvovaginitis (girls)	Dysuria from vulval irritation, discharge, poor hygiene, pinworm	Inspection, urine culture usually negative
Pinworm	Night perianal itch, vulval irritation	Sticky-tape test
Sexual abuse / STI	Discharge, genital injury, behaviour change, recurrent symptoms	Sensitive history, examination by trained staff, safeguarding
S. haematobium	Terminal haematuria, water contact	Midday urine for eggs
Acute glomerulonephritis	Coca-Cola urine, oedema, hypertension, recent sore throat/skin infection	RBC casts, ASO, complement
Urolithiasis	Colicky loin pain, haematuria	USS
Diabetes mellitus	Polyuria, new wetting, thirst, weight loss	Urine glucose and ketones, blood glucose
Kawasaki disease	Sterile pyuria with fever ≥5 days, conjunctivitis, rash	Clinical

Investigations

The full table with Ugandan availability and the NICE imaging schedule are in **Clinical workup, Step 4**. Remember:

1. **Collect urine before antibiotics** whenever it does not delay treatment of a sick child.
2. **Send culture** for: under 3 months, suspected upper UTI, high/intermediate risk of serious illness, discordant dipstick and clinical picture, recurrent or atypical UTI, failure to respond.
3. **Process within 4 hours** or refrigerate (boric acid containers per manufacturer). UCG 2023: get the specimen to the lab within 2 hours, or refrigerate at 4 °C for no more than 24 hours.
4. **Don't use CRP alone** to decide upper vs lower UTI (NICE).
5. **Don't image every child**: follow the age-and-type schedule.

Management

A. Management algorithm

1. **Sick or under 3 months?** → admit, resuscitate, blood culture (and LP if indicated), urine by catheter/SPA, **IV antibiotics** now.
2. **Over 3 months, well, upper UTI → oral antibiotics 7–10 days** (UCG: 10 days) if tolerating orally; IV for 24–72 h if vomiting, dehydrated or seriously ill, then oral.
3. **Over 3 months, lower UTI → oral antibiotics 3 days** (NICE) or per UCG.
4. **Supportive care**: fluids (oral or IV), **paracetamol 15 mg/kg** for fever and pain (avoid NSAIDs if dehydrated or renal impairment), encourage frequent voiding.
5. **Review at 48 h**: culture result; if **no improvement**, it is atypical UTI: **ultrasound now** (obstruction, abscess), check sensitivities, consider resistant organisms.
6. **Plan imaging** by age and UTI type (NICE table).
7. **Prevent recurrence**: treat constipation and BBD, fluids, hygiene; prophylaxis only for selected children.
8. **Follow up**: BP, growth, imaging results in writing, nephrology referral for scars, high-grade VUR or abnormal renal function.

B. Infants under 3 months

Drug	Dose	Route	Frequency	Duration	Notes
Ampicillin	**50 mg/kg**	IV/IM	**12-hourly in week 1**, 8-hourly in weeks 2–4, 6-hourly after 1 month	**10–14 days** in total (IV until afebrile 24–48 h and improving, then oral per culture)	Covers *Enterococcus*, GBS
plus Gentamicin	**5 mg/kg** (term, week 1), **3 mg/kg** (LBW/preterm, week 1); **7.5 mg/kg** after week 1 (WHO PB)	IV/IM	**Once daily**	Usually 5–7 days	Check creatinine; avoid prolonged use. UCG neonatal regimen uses 2.5 mg/kg 12-hourly
or Cefotaxime (1–3 months; or neonate)	**50 mg/kg**	IV	8-hourly (neonate 12-hourly in week 1)	As above	Use with ampicillin to cover *Enterococcus*

Drug	Dose	Route	Frequency	Duration	Notes
or Ceftriaxone (over 1 month, not jaundiced)	**50–80 mg/kg**	IV/IM	Once daily	As above	Not in jaundiced/preterm neonates or with calcium fluids

Do a **blood culture**, consider an **LP** (fever in a neonate), and image with **ultrasound** before discharge or within 6 weeks.

C. Children over 3 months: upper UTI (pyelonephritis)

Drug	Dose	Route	Frequency	Duration	Notes
Ceftriaxone (if vomiting, dehydrated or seriously ill)	**50–80 mg/kg** (UCG; max 2 g)	IV/IM	**Once daily**	2–4 days, then oral to complete **10 days**	First choice parenteral in Uganda
Gentamicin (alternative)	**7.5 mg/kg** (UCG/WHO PB)	IV/IM	Once daily	2–5 days, then oral	± ampicillin 50 mg/kg 6-hourly (WHO PB; UCG 25 mg/kg) for *Enterococcus*; avoid with raised creatinine
Cefixime (oral)	**8 mg/kg** (UCG: 16 mg/kg on day 1, then 8 mg/kg); max 400 mg	Oral	Once daily	To complete **7–10 days** (UCG 10)	Good oral step-down
Cefalexin (oral)	**25 mg/kg** (max 1 g)	Oral	8–12-hourly	7–10 days	NICE first-line oral (local resistance)
Co-amoxiclav (oral)	Amoxicillin component **about 10–15 mg/kg**	Oral	8-hourly	7–10 days	Only if culture shows susceptibility (NICE)
Ciprofloxacin (oral, second line/resistant)	**10–20 mg/kg** (max 750 mg)	Oral	12-hourly	7–10 days	For resistant organisms on culture; specialist advice
Meropenem (ESBL, severe)	**20 mg/kg** (max 1 g)	IV	8-hourly	Per culture	Referral hospitals; antimicrobial stewardship

D. Children over 3 months: lower UTI (cystitis)

Drug	Dose	Route	Frequency	Duration	Notes
Nitrofurantoin	UCG 2023: **3 mg/kg/day** in **4 divided doses**; Medscape range 5–7 mg/kg/day	Oral, after food	6-hourly	**UCG 7 days**; NICE **3 days**	**Cystitis only** (no kidney levels); avoid in G6PD deficiency, renal impairment, under 3 months
Trimethoprim	4 mg/kg (max 200 mg)	Oral	12-hourly	3 days	High resistance in Uganda: use only if sensitive

Drug	Dose	Route	Frequency	Duration	Notes
Cefalexin	12.5–25 mg/kg	Oral	12-hourly (or 8-hourly)	3 days	Alternative
Amoxicillin	UCG: 125–250 mg 8-hourly	Oral	8-hourly	7 days (UCG)	**Only if culture-sensitive**: high resistance

UGANDA CONTEXT

UCG 2023 vs NICE. UCG lists **nitrofurantoin 3 mg/kg/day in 4 doses for 7 days** (second line amoxicillin) for cystitis and **ceftriaxone 50–80 mg/kg once daily then cefixime to complete 10 days** (alternative gentamicin ± ampicillin) for pyelonephritis, and refers children with recurrent infection for investigation. NICE uses **3 days** for cystitis and **7–10 days of oral treatment** (cefalexin or co-amoxiclav) for pyelonephritis in children over 3 months. On Ugandan wards you will see the UCG regimens; choose by **culture sensitivity** whenever it is available, because resistance to amoxicillin and co-trimoxazole is high.

DRUG DOSING

Do not use **nitrofurantoin for pyelonephritis** (no renal tissue levels). **Gentamicin**: once-daily dosing, check creatinine, avoid with dehydration or renal impairment and limit duration (ototoxicity, nephrotoxicity). **Ceftriaxone** is avoided in jaundiced or preterm neonates and must not be given with calcium-containing fluids in neonates.

E. Recurrent UTI and VUR: prevention

Measure	Details
Treat constipation	Lactulose (about 0.5–1 mL/kg twice daily) or PEG, fluids, fibre, toilet routine
Voiding advice	Regular timed voiding every 2–3 hours, double voiding, relaxed posture (feet supported), **don't hold urine**, access to clean school toilets (NICE)
Fluids and hygiene	Adequate water intake; front-to-back wiping; avoid bubble baths/irritants
Circumcision	Consider in **infant boys with recurrent UTI or high-grade VUR** (urology advice)
Antibiotic prophylaxis	**Not routine after a first UTI** (NICE). Consider for **recurrent UTI**, **high-grade VUR (III–V)**, infants with hydronephrosis or awaiting surgery: **trimethoprim 2 mg/kg at night** (max 100 mg), **nitrofurantoin 1 mg/kg at night** (max 50 mg), or **cefalexin 10–12.5 mg/kg at night** (infants under 6 weeks; avoid nitrofurantoin and sulfonamides under 6 weeks)
Evidence (RIVUR, NEJM 2014)	In children 2–71 months with VUR grade I–IV after a febrile UTI, **TMP–SMX prophylaxis roughly halved recurrent UTIs** but **did not reduce renal scarring**, and increased resistant infections. So prophylaxis is a judgement for higher-risk children, not automatic
VUR surgery	Endoscopic subureteric injection (e.g. dextranomer/hyaluronic acid) or **ureteric reimplantation** for high-grade VUR with breakthrough infections or new scars (referral centres)
Posterior urethral valves	**Catheterise** (feeding tube 6–8 Fr) to drain the bladder, treat infection, monitor creatinine and electrolytes (post-obstructive diuresis), **urgent paediatric urology** for valve ablation; lifelong renal follow-up

F. Follow-up

- **Do not routinely re-test urine** in a child who is well after treatment (NICE).
- **Give imaging results in writing** to the parents.
- **Children with renal scars** (DMSA): **height, weight, BP and urine protein** at follow-up; **bilateral abnormalities, raised BP, proteinuria or impaired function → paediatric nephrologist** (NICE).
- **Teach parents** to seek care early for fever or urinary symptoms and to ask for a **urine test**.

G. Setting-specific plan

Setting	Action
HC II / HC III	Urine dipstick for febrile children without a focus (and RDT); clean-catch technique; **oral antibiotics for well children over 3 months** (cefixime if stocked; nitrofurantoin for cystitis); **refer infants under 3 months, sick children, boys with poor stream or palpable bladder**; give pre-referral IM ceftriaxone or gentamicin for sick children
HC IV	Admission, IV ceftriaxone/gentamicin, urine microscopy, ultrasound where available, catheterisation
RRH / NRH	Culture and sensitivity, blood culture, LP, MCUG (Mulago and selected centres), DMSA (Mulago), paediatric nephrology and urology, management of ESBL infections

Complications

Timing	Complications
Acute	**Urosepsis/septic shock** (infants), **meningitis** (neonates), dehydration, **acute kidney injury**, hyponatraemia/hyperkalaemia (pseudo-hypoaldosteronism in infants with obstruction), renal or perinephric **abscess**, **pyonephrosis** (infected obstructed kidney), febrile seizures
Long-term	**Renal scarring** (reflux nephropathy), **hypertension**, **proteinuria**, **chronic kidney disease**, poor growth, recurrent UTI; complications in later pregnancy (pre-eclampsia, pyelonephritis) for women with scarred kidneys
From treatment	Antibiotic resistance (prophylaxis), gentamicin nephro- and ototoxicity, *C. difficile*, allergy

Ward-round & exam pearls

Model case presentation

> "Baby M, a 4-month-old boy from Mukono, **weight 6.1 kg (weight-for-age Z −0.9), length 62 cm (length-for-age Z −0.5)**, **exclusively breastfed**, **immunised for age** (BCG scar, OPV/pentavalent/PCV/rotavirus 1–3 at 6, 10, 14 weeks), mother **HIV-negative** in this pregnancy, admitted yesterday with 2 days of fever, vomiting and irritability. The mother says he 'dribbles' urine and strains. Antenatal scan at 32 weeks: 'kidneys swollen' (not followed up). On admission: temperature 39.6 °C, HR 172, RR 48, CRT 2 s, irritable, fontanelle flat, **palpable bladder 4 cm above the pubis**, both kidneys ballotable; normal genitalia, spine normal. RDT negative. **Catheter urine: leucocytes 3+, nitrite +, >50 WBC/mm³, Gram-negative rods**; creatinine **98 µmol/L** (raised for age), Na 131, K 5.8 mmol/L. Ultrasound: **bilateral hydroureteronephrosis, thick-walled trabeculated bladder, dilated posterior urethra**. **Assessment: febrile UTI with sepsis in an infant boy, with atypical features (poor stream, bladder mass, raised creatinine) strongly suggesting posterior urethral valves. Management so far:** urethral catheter (6 Fr feeding tube) left on free drainage, **ampicillin 300 mg IV 6-hourly + gentamicin** avoided because of the raised creatinine (switched to

> **ceftriaxone 450 mg IV daily**), IV fluids with monitoring for post-obstructive diuresis, paracetamol. **Plan:** culture results; daily creatinine and electrolytes; **MCUG** once infection is controlled; **paediatric urology referral** for valve ablation; nephrology follow-up for CKD; BP monitoring; continue breastfeeding."

Likely viva questions

1. *How do you diagnose UTI in a 6-month-old?* Properly collected urine (clean catch/pad, or catheter/SPA if sick), dipstick or microscopy, and culture; fever with bacteriuria = upper UTI.
2. *Which dipstick result is most specific?* Nitrite (but insensitive in infants).
3. *Define atypical UTI.* Seriously ill, poor urine flow, abdominal or bladder mass, raised creatinine, septicaemia, no response within 48 h, non-*E. coli* organism.
4. *Imaging after a first typical UTI in a 2-year-old who responded in 48 h?* None (NICE); AAP would scan the kidneys after a first febrile UTI in 2–24 months.
5. *What is the commonest organism? E. coli* (80–90%).
6. *Grade V VUR?* Gross dilatation and tortuosity of the ureter and pelvicalyceal system with loss of papillary impressions.
7. *What did RIVUR show?* Prophylaxis halved recurrences in VUR I–IV but did not reduce scarring.

OSCE tips: in a counselling station on clean-catch collection, explain hand washing, cleaning the genitals with water (no antiseptic), catching the midstream in a sterile pot, closing it and bringing it within an hour (or refrigerating). For a urine dipstick station: check the expiry date, dip briefly, tap off excess, hold horizontally, **read each pad at the right time**, and interpret leucocytes and nitrites.

Mnemonics & memory aids

- **Atypical UTI "SPARSE-N"**: **S**eriously ill, **P**oor urine flow, **A**bdominal/bladder mass, **R**aised creatinine, **S**epticaemia, **E**ffect of antibiotics absent at 48 h, **N**on-*E. coli*.
- **Recurrent UTI "2-1+1-3"**: 2 upper; 1 upper + 1 lower; 3 lower.
- **Culture thresholds "SPA any, Cath 10, Clean 100"**: any growth / ≥10,000 / ≥100,000 CFU/mL (AAP catheter ≥50,000).
- **Nitrite = "Not Infant-friendly"**: specific but often negative in infants (frequent voiding).
- **Imaging "USS for the young and the atypical; DMSA for scars; MCUG for babies"**.
- **VUR grades "Ureter, Up to calyces, a bit Dilated, very Dilated and Tortuous, Gross"** (I–V).

High-yield summary

Numbers & doses to know

Item	Value
UTI in febrile 2–24 month olds without a source	**~5–7%**
E. coli share	**80–90%**
SPA threshold	**Any growth**
Catheter threshold	**≥10,000 CFU/mL** (AAP **≥50,000**)
Clean catch threshold	**≥100,000 CFU/mL** single organism
Pyuria	**≥10 WBC/mm³** (WHO PB **>5 WBC/hpf**)
Nitrite	Specific **>90%**; sensitivity **~50%**

Item	Value
Leucocyte esterase	Sensitivity **~80%**
Process urine	Within **4 h** (or refrigerate; UCG within 2 h or ≤24 h at 4 °C)
Upper UTI definition	Bacteriuria + **fever ≥38 °C** or loin pain
Under 3 months	**IV** ampicillin + gentamicin (neonate) / cefotaxime or ceftriaxone + ampicillin; 10–14 days
Ceftriaxone (UCG)	**50–80 mg/kg** IV/IM once daily (max 2 g)
Gentamicin	**7.5 mg/kg** once daily (child); **5 mg/kg** (term neonate week 1), **3 mg/kg** (LBW week 1)
Ampicillin	**50 mg/kg** 6-hourly (child)
Cefixime	**8 mg/kg** once daily (UCG 16 mg/kg day 1)
Cefalexin	**25 mg/kg** 8–12-hourly (upper); 12.5–25 mg/kg 12-hourly (lower)
Nitrofurantoin	UCG **3 mg/kg/day ÷ 4** × 7 days (cystitis only)
Upper UTI duration	**7–10 days** (UCG 10)
Lower UTI duration	**3 days** (NICE); 7 days (UCG)
Prophylaxis	Trimethoprim **2 mg/kg** nocte; nitrofurantoin **1 mg/kg** nocte
DMSA timing	**4–6 months** after infection
MCUG antibiotic cover	**3 days**, MCUG on day 2
Recurrent UTI	≥2 upper; 1 upper + ≥1 lower; ≥3 lower

Classic exam traps

- **"The bag urine grew 10^5 mixed organisms: start antibiotics and order an MCUG."** Wrong: likely contamination; recollect properly.
- **"Negative nitrite rules out UTI in an infant."** Wrong: infants void too frequently for nitrite to form.
- **"CRP of 120 means pyelonephritis."** NICE: don't use CRP alone to localise.
- **"Nitrofurantoin for a febrile 2-year-old."** Wrong: no renal levels; use cefixime/cefalexin/ceftriaxone.
- **"Every child needs an MCUG after a first UTI."** Wrong: MCUG is for atypical/recurrent UTI under 6 months or specific features.
- **"Prophylaxis for every child after a first UTI."** Wrong (NICE); RIVUR showed no effect on scarring.
- **"Amoxicillin is a good empirical choice in Uganda."** Wrong: high resistance; use culture-guided therapy.
- **"Treat asymptomatic bacteriuria."** Wrong in children (NICE).

Questions seniors ask

Why did you send this urine for culture when the dipstick was positive? The child is under 3 years with suspected upper UTI (and/or high risk of serious illness); culture confirms the diagnosis, guides antibiotics in a high-resistance setting and defines future recurrences.

When would you give IV rather than oral antibiotics? Infants under 3 months, children who are seriously ill or septic, vomiting or unable to take oral medicine, or dehydrated; switch to oral when improving (usually 24–72 h).

This 8-month-old boy has his second febrile UTI. What imaging will you request? Recurrent UTI at 6 months–3 years (NICE): ultrasound within 6 weeks and DMSA at 4–6 months; consider MCUG if the ultrasound shows dilatation, poor flow, a non-*E. coli* organism or a family history of VUR. AAP: VCUG after a second febrile UTI.

Why does a palpable bladder in an infant boy worry you? It suggests bladder outflow obstruction from **posterior urethral valves**; he needs catheter drainage, creatinine and electrolytes, urgent ultrasound and MCUG, and paediatric urology.

Should this girl with VUR grade II get prophylaxis? Probably not after a single UTI: low-grade VUR usually resolves; focus on constipation and voiding habits; consider prophylaxis if UTIs recur (RIVUR: fewer recurrences, no less scarring).

How do you follow up a child with renal scarring on DMSA? Regular height, weight, **BP** and **urine protein**; refer to nephrology if scarring is bilateral, BP is raised, there is proteinuria or renal function is impaired.

SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Which febrile children need a urine test?	Any infant or young child with **fever and no obvious source** (even after a negative malaria test), all febrile infants under 3 months, and children with urinary symptoms. Half of infants with UTI have fever only.
2	Preferred urine collection methods in children (NICE)?	**Clean catch** first; if not possible, **urine collection pad** (never cotton wool/gauze); if not possible or very sick, **catheter** or **ultrasound-guided SPA**. Bag samples only to rule out (negative is useful).
3	Significant culture growth by method?	**SPA: any growth**; **catheter: ≥10,000 CFU/mL** (AAP ≥50,000 in febrile 2–24 month olds); **clean catch: ≥100,000 CFU/mL** of a single organism. Mixed growth = contamination.
4	Nitrite and leucocyte esterase: sensitivity and specificity?	**Nitrite**: specific (>90%) but **sensitivity ~50%** (needs ~4 h bladder dwell and nitrate-reducing organisms; often negative in infants). **LE**: sensitivity ~80%, less specific.
5	NICE dipstick strategy for children 3 months to 3 years?	**LE or nitrite (or both) positive → send culture and give antibiotics.** Both negative → no antibiotics and no culture unless other criteria apply. Under 3 months: urgent microscopy and culture instead.
6	NICE dipstick strategy for children ≥3 years?	LE+ nitrite+ → treat (culture if high/intermediate risk or previous UTI); LE− nitrite+ → treat and culture; LE+ nitrite− → culture, treat only with good clinical evidence; both negative → not UTI.
7	Definition of upper UTI (NICE)?	Bacteriuria with **fever ≥38 °C**, or bacteriuria with fever <38 °C and **loin pain/tenderness**. Bacteriuria without systemic features = lower UTI.
8	Atypical UTI (NICE)?	**Seriously ill; poor urine flow; abdominal or bladder mass; raised creatinine; septicaemia; failure to respond within 48 h; non-*E. coli* organism.** (SPARSE-N)
9	Recurrent UTI (NICE)?	**≥2 upper UTIs**, or **1 upper + ≥1 lower**, or **≥3 lower UTIs**.
10	Commonest organisms in childhood UTI?	***E. coli* (80–90%)**; *Klebsiella*, *Proteus* (boys, stones), *Enterococcus* (infants, abnormal tracts), *Pseudomonas* (catheters), *S. saprophyticus* (adolescent girls). ESBL producers increasing.
11	Management of UTI in an infant under 3 months?	**Admit; IV antibiotics**: neonate **ampicillin 50 mg/kg + gentamicin** (5 mg/kg term week 1; 3 mg/kg LBW); 1–3 months cefotaxime/ceftriaxone + ampicillin; **blood culture**, consider **LP**; 10–14 days total; ultrasound.
12	UCG 2023 treatment of pyelonephritis in children?	**Ceftriaxone 50–80 mg/kg IV once daily**, then **cefixime** (16 mg/kg day 1 then 8 mg/kg/day) to complete **10 days**; alternative **gentamicin 7.5 mg/kg once daily** ± ampicillin. Refer recurrent infections.
13	UCG 2023 treatment of cystitis in children?	**Nitrofurantoin 3 mg/kg/day in 4 divided doses for 7 days** (after meals); second line amoxicillin (only if sensitive). NICE: 3 days for children over 3 months.
14	Why not use nitrofurantoin for pyelonephritis?	It achieves **urinary but not renal tissue or blood levels**, so it cannot treat kidney infection or bacteraemia. Use it only for cystitis.
15	NICE imaging for a child under 6 months with a first typical UTI responding within 48 h?	**Ultrasound within 6 weeks**; if abnormal, consider MCUG. No DMSA or routine MCUG.

#	Question	Answer
16	NICE imaging for atypical UTI under 6 months?	**Ultrasound during the acute infection**, **DMSA 4–6 months** later, and **MCUG**.
17	NICE imaging for a child 6 months–3 years with recurrent UTI?	**Ultrasound within 6 weeks** and **DMSA at 4–6 months**; MCUG not routine (consider if dilatation, poor flow, non-*E. coli*, family history of VUR).
18	AAP 2011 imaging for febrile UTI at 2–24 months?	**Renal and bladder ultrasound after the first febrile UTI**; **VCUG only if the ultrasound is abnormal** (hydronephrosis, scarring, suspected high-grade VUR/obstruction) or after a **second** febrile UTI.
19	VUR grades I–V?	**I** ureter only; **II** ureter + pelvis/calyces, no dilatation; **III** mild–moderate dilatation; **IV** moderate dilatation and tortuosity, blunted fornices; **V** gross dilatation/tortuosity, loss of papillary impressions.
20	What did the RIVUR trial (2014) show?	In children 2–71 months with **VUR grade I–IV** after a febrile UTI, TMP–SMX prophylaxis **roughly halved recurrent UTIs** but **did not reduce renal scarring**, and increased resistant infections.
21	Prophylactic antibiotic doses?	**Trimethoprim 2 mg/kg at night** (max 100 mg) or **nitrofurantoin 1 mg/kg at night** (max 50 mg); **cefalexin ~10–12.5 mg/kg** at night under 6 weeks. Not routine after a first UTI.
22	Clues to posterior urethral valves?	**Infant boy** with UTI, **weak or dribbling stream**, straining, **palpable bladder**, bilateral hydronephrosis/antenatal hydronephrosis, raised creatinine. **Catheterise**, ultrasound, **MCUG**, urgent paediatric urology.
23	What is bladder and bowel dysfunction?	**Voiding dysfunction** (holding, infrequent voiding, incomplete emptying, wetting) with **constipation**; a major cause of recurrent UTI in school-age children. Treat constipation and use timed voiding.
24	When is a DMSA scan done and what does it show?	**4–6 months after** an atypical/recurrent UTI (per age, NICE); shows **renal parenchymal scars** and differential function. In Uganda: Mulago nuclear medicine only.
25	Follow-up for a child with renal scarring?	**Height, weight, blood pressure and urine protein**; refer to a paediatric nephrologist if bilateral scars, raised BP, proteinuria or impaired renal function (NICE).
26	How fast must urine reach the lab?	Culture within **4 hours** or refrigerate (NICE); UCG 2023: within **2 hours**, or refrigerate at 4 °C for no more than **24 hours**.
27	Should asymptomatic bacteriuria be treated in children?	**No** (NICE) — treatment does not prevent scarring and promotes resistance (exceptions such as pregnancy or before urological surgery).

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 10-month-old girl (8.4 kg) has had fever of 39.4 °C for 2 days without cough, diarrhoea or rash. She is alert and feeding, HR 150, RR 38, CRT 2 s. Ears, throat and chest are normal. Malaria RDT is negative. What is the MOST appropriate next step?

A Reassure: viral fever; paracetamol and review in 3 days
B Start oral amoxicillin empirically
C Repeat the malaria RDT tomorrow before any other test
D Attach a urine bag and send the sample for culture
E Collect a clean-catch (or pad) urine for dipstick, and send culture if positive

2. A 3-week-old boy (3.6 kg) presents with fever 38.6 °C, poor feeding, jaundice and lethargy. HR 182, CRT 3 s. Catheter urine: leucocytes 2+, nitrite negative, microscopy shows pyuria and Gram-negative rods. What is the BEST management?

A Oral cefixime and outpatient review
B Oral nitrofurantoin for 7 days
C Admit; blood culture, consider LP; IV ampicillin plus gentamicin (or cefotaxime plus ampicillin); fluids; 10–14 days of treatment
D Wait for urine culture before starting antibiotics
E IV ceftriaxone running with Ringer's lactate

3. A 2-year-old girl (12 kg) has fever 39 °C, vomiting everything and right loin tenderness. Catheter urine: leucocytes 3+, nitrite positive. HR 150, CRT 2 s, mildly dehydrated. Creatinine normal. Which initial antibiotic follows UCG 2023?

A Ceftriaxone 600–960 mg (50–80 mg/kg) IV once daily, switching to oral cefixime when she can take it, for 10 days in total
B Oral amoxicillin 125 mg 8-hourly for 7 days
C Oral nitrofurantoin 9 mg 6-hourly for 7 days
D Oral co-trimoxazole for 3 days
E IV ceftriaxone single dose then stop

4. A 7-year-old girl (22 kg) has 2 days of dysuria and frequency without fever or loin pain. Midstream urine dipstick: leucocytes 2+, nitrite positive. She is otherwise well. Which treatment is MOST appropriate under UCG 2023?

A IV ceftriaxone for 10 days
B Gentamicin 7.5 mg/kg IM once daily for 5 days
C Oral ciprofloxacin for 14 days
D Oral nitrofurantoin about 3 mg/kg/day in 4 divided doses (about 16 mg 6-hourly) for 7 days, with fluids and review of bowel and voiding habits
E No treatment: cystitis resolves spontaneously

5. A 14-month-old boy (10 kg) has his urine sent from a bag attached for 2 hours. The culture reports 'mixed growth of three organisms, >10^5 CFU/mL'. He is afebrile today and well. What should you do?

A Start oral antibiotics for 10 days and request an MCUG
B Regard it as probable contamination; if UTI is still suspected, recollect by clean catch or catheter
C Start prophylactic trimethoprim
D Request a DMSA scan
E Diagnose recurrent UTI

6. A 4-month-old boy (6.1 kg) with febrile UTI has a weak dribbling stream, a bladder palpable 4 cm above the pubis, creatinine 98 µmol/L and bilateral hydronephrosis on ultrasound. What is the MOST important next step after starting antibiotics?

A DMSA scan in 4–6 months only
B Prophylactic trimethoprim and discharge
C Oral fluids and repeat ultrasound in 6 weeks
D Circumcision this week
E Drain the bladder with a urethral catheter (feeding tube), monitor creatinine and electrolytes, and arrange MCUG and urgent paediatric urology referral

7. A 2-year-old girl (12 kg) had her first febrile UTI with E. coli, confirmed on a catheter sample. She responded to treatment within 48 hours and is now well. Normal stream, no masses, normal creatinine. According to NICE NG224, what imaging is required?

A Ultrasound during the acute infection, DMSA at 4–6 months and MCUG
B MCUG only
C No routine imaging
D DMSA scan only
E CT urogram

8. A 15-month-old (9.5 kg) has fever 38.9 °C for a day. Clean-catch urine dipstick: leucocyte esterase negative, nitrite negative. She has coryza and a red throat; RDT negative. What is the BEST interpretation?

A UTI is unlikely: do not give antibiotics for UTI or send a culture (unless other criteria apply); manage the likely viral URTI and review
B Start antibiotics for UTI and send culture
C Repeat the dipstick on a bag sample
D Send for a DMSA scan
E Give nitrofurantoin for 3 days to be safe

9. A 6-year-old girl (20 kg) has had three episodes of cystitis in 8 months. She wets her pants in the day, opens her bowels every 4–5 days with hard stools, and avoids the school latrine. Examination shows faecal loading; BP normal. What is the MOST important intervention?

A Long-term ciprofloxacin prophylaxis
B Urgent MCUG and ureteric reimplantation
C Cranberry juice only
D Treat constipation (e.g. lactulose or PEG), timed voiding every 2–3 hours, adequate fluids and access to clean toilets, plus imaging for recurrent UTI
E Restrict fluids to reduce wetting

10. A 3-year-old boy (14 kg) with pyelonephritis has received IV ceftriaxone for 48 hours, but his fever persists at 39.5 °C and he has worsening right loin pain. Culture shows Klebsiella resistant to ceftriaxone. What is the BEST next step?

A Continue ceftriaxone for another 5 days

B Urgent renal ultrasound to look for obstruction or abscess, and change antibiotics according to sensitivities (e.g. meropenem for an ESBL)

C Add oral amoxicillin

D Stop antibiotics; it is a viral fever

E Discharge on nitrofurantoin

11. Which urine collection method is the gold standard for diagnosing UTI in a sick 2-month-old?

A Urine bag

B Cotton wool ball in the nappy

C Midstream sample from the nappy

D Urine from a 24-hour collection

E Ultrasound-guided suprapubic aspiration

12. A 9-month-old boy (8.5 kg) had his second febrile E. coli UTI. Ultrasound shows a dilated left renal pelvis and ureter. MCUG shows left VUR with moderate dilatation and tortuosity of the ureter and blunting of the fornices, with papillary impressions preserved. What grade is this, and which management is reasonable?

A Grade II; no follow-up needed

B Grade V; immediate nephrectomy

C Grade IV; consider continuous antibiotic prophylaxis (e.g. trimethoprim 2 mg/kg at night), treat constipation, DMSA in 4–6 months and paediatric urology/nephrology follow-up

D Grade I; stop all follow-up

E Grade III; ureteric reimplantation this week

13. A 3-year-old (14 kg) with a febrile UTI has a CRP of 140 mg/L. A student says this proves pyelonephritis and the child therefore needs a DMSA scan. What is the BEST response?

A NICE advises against using CRP alone to separate upper from lower UTI; diagnose upper UTI clinically (fever ≥38 °C or loin pain with bacteriuria) and image according to age and type of UTI

B Correct: CRP above 100 mg/L confirms pyelonephritis

C CRP should be repeated daily to guide imaging

D CRP below 20 mg/L excludes UTI

E A raised CRP means MCUG is required

14. An 11-year-old boy (30 kg) from a riverside village in eastern Uganda reports blood at the end of urination for 2 months, without dysuria or fever. Dipstick: blood 3+, leucocytes negative, nitrite negative. Which diagnosis is MOST likely?

A Acute pyelonephritis

B Cystitis from E. coli

C Bladder and bowel dysfunction

D Urogenital schistosomiasis (S. haematobium)

E Nephrotic syndrome

15. Which statement about antibiotic prophylaxis after childhood UTI is CORRECT?

- A Every child should receive prophylaxis after a first UTI
- B In RIVUR, prophylaxis reduced recurrent UTIs in children with VUR but did not reduce renal scarring
- C Nitrofurantoin prophylaxis is safe from birth
- D Prophylaxis eliminates the need to treat constipation
- E Ciprofloxacin is the first-line prophylactic agent in infants

Answers & explanations

1. E · Collect a clean-catch (or pad) urine for dipstick, and send culture if positive

E is correct: fever without a source in a child under 2 with a negative RDT needs a **urine test**; a **clean-catch** (or pad) sample is preferred, with **dipstick** (LE/nitrite) and **culture if positive** (NICE). A is wrong: UTI is found in about 5–7% of such children and can scar the kidney. B is wrong: amoxicillin is a poor empirical choice in Uganda and treating without a sample prevents diagnosis. C is wrong: a negative RDT with no symptoms of malaria should prompt a search for other causes now. D is wrong: bag cultures have a very high false-positive rate; a positive bag culture is unreliable.

2. C · Admit; blood culture, consider LP; IV ampicillin plus gentamicin (or cefotaxime plus ampicillin); fluids; 10–14 days of treatment

C is correct: a febrile, lethargic neonate with UTI has presumed **urosepsis**; admit, take **blood culture**, consider **LP**, and give **IV ampicillin + gentamicin** (or cefotaxime + ampicillin) with fluid support; treat for 10–14 days. A is wrong: infants under 3 months need parenteral treatment in hospital (NICE). B is wrong: nitrofurantoin doesn't treat sepsis or pyelonephritis and is avoided in neonates. D is wrong: treatment must not be delayed in a septic infant. E is wrong: ceftriaxone is avoided in a **jaundiced neonate** and must not be given with **calcium-containing fluids**. A negative nitrite does not exclude UTI in infants.

3. A · Ceftriaxone 600–960 mg (50–80 mg/kg) IV once daily, switching to oral cefixime when she can take it, for 10 days in total

A is correct: this is **pyelonephritis** in a child who is **vomiting everything**, so parenteral treatment is needed: UCG 2023 gives **ceftriaxone 50–80 mg/kg once daily** (12 kg → 600–960 mg), then oral **cefixime** to complete **10 days**. B is wrong: oral treatment is not possible while she vomits everything, and amoxicillin resistance is high. C is wrong: nitrofurantoin does not reach the kidney. D is wrong: co-trimoxazole resistance is high and 3 days is a cystitis course. E is wrong: a single dose is inadequate for pyelonephritis.

4. D · Oral nitrofurantoin about 3 mg/kg/day in 4 divided doses (about 16 mg 6-hourly) for 7 days, with fluids and review of bowel and voiding habits

D is correct: this is **lower UTI (cystitis)**; UCG 2023 gives **nitrofurantoin 3 mg/kg/day in 4 doses for 7 days** (NICE: 3 days); also address constipation and voiding habits. A is wrong: IV treatment for 10 days is for pyelonephritis in sick children. B is wrong: injectable aminoglycoside is unnecessary for uncomplicated cystitis. C is wrong: fluoroquinolones are second line for resistant infections, and 14 days is excessive. E is wrong: symptomatic bacterial cystitis should be treated.

5. B · Regard it as probable contamination; if UTI is still suspected, recollect by clean catch or catheter

B is correct: **mixed growth from a bag sample** is almost always **contamination**; if UTI is still suspected, collect a reliable sample (clean catch, catheter or SPA). A is wrong: treating and imaging on a contaminated bag culture leads to unnecessary antibiotics and invasive tests. C is wrong: there is no confirmed UTI, let alone recurrence. D is wrong: DMSA is for confirmed atypical or recurrent UTI. E is wrong: recurrence requires confirmed episodes.

6. E · Drain the bladder with a urethral catheter (feeding tube), monitor creatinine and electrolytes, and arrange MCUG and urgent paediatric urology referral

E is correct: an infant boy with poor stream, a **palpable bladder**, **bilateral hydronephrosis** and **raised creatinine** has **posterior urethral valves** until proven otherwise; **catheter drainage**, electrolyte/creatinine monitoring (post-obstructive diuresis, hyperkalaemia), **MCUG** and **paediatric urology** for valve ablation are urgent. A is wrong: DMSA does not address obstruction. B is wrong: prophylaxis without relieving obstruction risks further renal damage. C is wrong: waiting 6 weeks with obstruction and rising creatinine is unsafe. D is wrong: circumcision does not relieve urethral valve obstruction.

7. C · No routine imaging

C is correct: NICE does **not** routinely image children **over 6 months** with a **first typical UTI** that responds within 48 h (table 5). A is wrong: that schedule is for **atypical** UTI in infants under 6 months. B is wrong: MCUG is not routine at 6 months–3 years. D is wrong: DMSA is for atypical or recurrent UTI. E is wrong: CT exposes the child to radiation and is not part of UTI imaging. (Note: the AAP would advise renal and bladder ultrasound after a first febrile UTI at 2–24 months, a well-known difference.)

8. A · UTI is unlikely: do not give antibiotics for UTI or send a culture (unless other criteria apply); manage the likely viral URTI and review

A is correct: in a child aged 3 months to 3 years, **both LE and nitrite negative** makes UTI unlikely (NICE): no antibiotics, no culture unless other criteria (e.g. high risk of serious illness), and look for another source (here a viral URTI). B is wrong: treating with negative dipstick and an alternative source is overtreatment. C is wrong: a bag sample is less reliable than the clean catch already done. D is wrong: there is no confirmed UTI. E is wrong: unnecessary antibiotics promote resistance.

9. D · Treat constipation (e.g. lactulose or PEG), timed voiding every 2–3 hours, adequate fluids and access to clean toilets, plus imaging for recurrent UTI

D is correct: recurrent cystitis with daytime wetting, constipation and holding is **bladder and bowel dysfunction**; treating **constipation** and **voiding habits** is the key intervention, and NICE recommends **ultrasound within 6 weeks and DMSA** for recurrent UTI at ≥3 years. A is wrong: fluoroquinolone prophylaxis is not appropriate and does not fix the cause. B is wrong: surgery without evidence of high-grade VUR is inappropriate. C is wrong: cranberry has weak evidence and ignores the cause. E is wrong: fluid restriction concentrates urine and worsens infection risk.

10. B · Urgent renal ultrasound to look for obstruction or abscess, and change antibiotics according to sensitivities (e.g. meropenem for an ESBL)

B is correct: failure to respond within 48 h and a **non-E. coli (resistant)** organism make this an **atypical UTI**; do an **ultrasound during the acute infection** (obstruction, pyonephrosis, abscess) and change to an antibiotic the organism is sensitive to (e.g. **meropenem 20 mg/kg 8-hourly** for ESBL). A is wrong: continuing an ineffective drug delays cure. C is wrong: amoxicillin will not cover a ceftriaxone-resistant Klebsiella. D is wrong: he has culture-proven bacterial infection. E is wrong: nitrofurantoin does not treat pyelonephritis.

11. E · Ultrasound-guided suprapubic aspiration

E is correct: **ultrasound-guided SPA** is the gold standard (any growth is significant); a catheter sample is the main alternative. A is wrong: bag samples are frequently contaminated. B is wrong: NICE says never use cotton wool, gauze or sanitary towels. C is wrong: urine from a nappy is contaminated. D is wrong: 24-hour collections are not used to diagnose UTI.

12. C · Grade IV; consider continuous antibiotic prophylaxis (e.g. trimethoprim 2 mg/kg at night), treat constipation, DMSA in 4–6 months and paediatric urology/nephrology follow-up

C is correct: **moderate dilatation and tortuosity with blunted fornices but preserved papillary impressions = grade IV VUR**; with recurrent febrile UTI, **prophylaxis** (trimethoprim 2 mg/kg nocte) is reasonable (RIVUR: fewer recurrences), plus **DMSA** for scarring and specialist follow-up; circumcision may be discussed in an infant boy. A is wrong: grade II has no dilatation. B is wrong: grade V means gross dilatation with loss of papillary impressions, and nephrectomy is not indicated. D is wrong: grade I is reflux into the ureter only. E is wrong: grade III is mild–moderate dilatation, and surgery is not first line.

13. A · NICE advises against using CRP alone to separate upper from lower UTI; diagnose upper UTI clinically (fever ≥38 °C or loin pain with bacteriuria) and image according to age and type of UTI

A is correct: NICE NG224 says **do not use CRP alone** to differentiate upper from lower UTI; upper UTI is defined clinically (bacteriuria with fever ≥38 °C, or with loin pain), and imaging follows the age/atypical/recurrent tables. B is wrong: CRP overlaps between upper and lower UTI and is raised by other infections. C is wrong: daily CRP does not determine imaging. D is wrong: a low CRP does not exclude UTI. E is wrong: MCUG indications are based on age, atypical features and recurrence, not CRP.

14. D · Urogenital schistosomiasis (S. haematobium)

D is correct: **terminal haematuria** with river exposure and **no pyuria or nitrite** suggests **S. haematobium**; confirm with eggs (terminal spine) in a midday urine and treat with **praziquantel 40 mg/kg**. A is wrong: no fever, loin pain or pyuria. B is wrong: cystitis would give dysuria, leucocytes and often nitrites. C is wrong: BBD gives wetting and recurrent UTI, not isolated terminal haematuria. E is wrong: nephrotic syndrome causes heavy proteinuria and oedema, not terminal haematuria.

15. B · In RIVUR, prophylaxis reduced recurrent UTIs in children with VUR but did not reduce renal scarring

B is correct: the **RIVUR trial (NEJM 2014)** showed TMP–SMX prophylaxis **roughly halved recurrences** in children with VUR grade I–IV but **did not reduce scarring**, and increased resistance. A is wrong: NICE says **do not routinely give prophylaxis after a first UTI**. C is wrong: nitrofurantoin and sulfonamides are avoided under 6 weeks. D is wrong: treating constipation and voiding dysfunction remains essential. E is wrong: trimethoprim, nitrofurantoin or cefalexin (young infants) are used, not ciprofloxacin.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: A febrile infant boy with a palpable bladder

You are the intern at Mengo Hospital paediatric emergency. A 4-month-old boy from Mukono is brought with 2 days of fever, vomiting and irritability. His mother says he 'dribbles' urine and strains. An antenatal scan at 32 weeks said 'kidneys swollen' but was not followed up. Weight 6.1 kg.

1. What are your first steps?

> ETAT: airway, breathing (RR 48), **circulation** (HR 172, CRT 2 s: not shocked), irritable but responsive, **glucose** 4.4 mmol/L. RDT (negative). IV access: **blood culture, FBC, creatinine, electrolytes**. Collect urine by **catheter** (sick infant) before antibiotics if quick; start IV antibiotics promptly.

2. Catheter urine: leucocytes 3+, nitrite +, >50 WBC/mm3, Gram-negative rods. Examination: palpable bladder 4 cm above the pubis, ballotable kidneys. What is your assessment?

> **Febrile UTI (presumed pyelonephritis) with urosepsis risk** in an infant under 6 months, with **atypical features** (poor urine flow, bladder mass, likely raised creatinine) suggesting **posterior urethral valves**.

3. Creatinine 98 µmol/L, Na 131, K 5.8 mmol/L. Which antibiotics and doses?

> Because creatinine is raised, **avoid gentamicin**: **ceftriaxone 75 mg/kg ≈ 450 mg IV once daily** (not jaundiced) **plus ampicillin 50 mg/kg = 300 mg IV 6-hourly** for *Enterococcus* cover until culture returns; adjust to sensitivities; total 10–14 days. Paracetamol 15 mg/kg = 90 mg.

4. What immediate non-antibiotic measures and monitoring are needed?

> **Leave a urethral catheter (6–8 Fr feeding tube) on free drainage** to relieve obstruction; **strict input/output**; IV fluids to match output (watch for **post-obstructive diuresis**); **daily creatinine and electrolytes** (hyperkalaemia, hyponatraemia); BP; continue breastfeeding.

5. What imaging and referrals will you arrange?

> **Ultrasound now** (done: bilateral hydroureteronephrosis, thick trabeculated bladder, dilated posterior urethra); **MCUG** once infection is controlled (with antibiotic cover) to confirm valves and VUR; **urgent paediatric urology** (valve ablation, Mulago or another national centre); **paediatric nephrology** follow-up for CKD; **DMSA** at 4–6 months (Mulago).

6. What will you tell the mother about the long-term outlook?

> The valves can be treated with a small operation, but the kidneys may already be damaged; he needs **lifelong follow-up** of kidney function, BP, growth and bladder function, and early care for any fever (urine test each time). Antenatal hydronephrosis should always be followed up.

Take-away: An infant boy with UTI and a poor stream or palpable bladder has posterior urethral valves until proven otherwise: drain the bladder, avoid nephrotoxins, scan, and refer to paediatric urology.

Case 2: A toddler with fever and no focus: getting the sample right

At a HC IV in Wakiso, an 18-month-old girl (10.5 kg) has had fever 39.2 °C for 2 days and vomited twice. She is drinking, alert, with no cough, rash or diarrhoea. HR 148, RR 36, CRT 2 s. Ears and throat normal. RDT negative. Her mother asks for 'an injection for malaria'.

1. How do you explain your plan to the mother, and which test do you do?

> Explain that the malaria test is negative and that **urine infection is a common hidden cause** of fever at this age. Collect a **clean-catch** urine (sit her on the mother's lap without a nappy, clean genitals with water, catch urine in a sterile pot) or use a urine pad; dipstick it straight away.

2. Dipstick: leucocytes 2+, nitrite +. What next?

> LE and nitrite positive at 3 months–3 years → **treat as UTI and send culture** (NICE). With fever ≥38 °C this is **upper UTI**. She is drinking and not septic → **oral treatment** possible: **cefixime 8 mg/kg ≈ 84 mg once daily** (UCG gives 16 mg/kg on day 1) to complete **10 days**; paracetamol 15 mg/kg ≈ 160 mg 6-hourly; fluids. Send culture to the hub lab.

3. Two days later she still has fever 39 °C and is vomiting. What do you do?

> Failure to respond within 48 h = **atypical UTI**: admit; **IV ceftriaxone 50–80 mg/kg ≈ 525–840 mg once daily**; check culture/sensitivities and creatinine; **ultrasound now** (obstruction, abscess); reconsider the diagnosis (malaria repeat, other foci).

4. Culture grows E. coli sensitive to ceftriaxone; ultrasound is normal; she recovers. What imaging is needed per NICE?

> Atypical UTI at 6 months–3 years (table 5): **ultrasound during the acute infection** (done, normal) and **DMSA at 4–6 months**; **MCUG not routine** (consider only if dilatation, poor flow, non-*E. coli* or family history of VUR). In Uganda, DMSA requires referral to Mulago; discuss feasibility.

5. What advice do you give at discharge?

> Complete the antibiotic course; seek care **early** for any future fever and **ask for a urine test**; adequate fluids; avoid constipation; nappy hygiene (wipe front to back); results of investigations in writing; follow-up for DMSA. No routine prophylaxis after a single UTI.

Take-away: Fever without a source in a toddler means a urine test on a clean sample; treat by age and severity, and let non-response at 48 hours trigger reassessment and imaging.

Case 3: Recurrent cystitis in a school girl

A 6-year-old girl (20 kg) from Kampala is brought to the paediatric clinic with her third episode of dysuria and frequency in 8 months. She has started wetting her pants during the day. She passes hard stools every 4–5 days and avoids the school latrine. No fever. BP 98/62.

1. What else must you ask and examine?

> Fever or loin pain with past episodes (upper vs lower UTI), how previous diagnoses were made (culture?), **bowel habit**, voiding frequency and holding, fluid intake, vaginal discharge or itch (vulvovaginitis, pinworm), **sensitive questions about abuse**, family history of VUR. Examine: abdomen (**faecal loading**), loins, genitalia (inspection only, with chaperone), **spine** (dimple, tuft), lower-limb neurology, BP, growth.

2. Clean-catch dipstick: LE +, nitrite +. Culture grows E. coli >10^5 CFU/mL, sensitive to nitrofurantoin. Treat this episode.

> **Lower UTI**: **nitrofurantoin 3 mg/kg/day ÷ 4 = 15 mg 6-hourly** after meals for 7 days (UCG; NICE 3 days); fluids; paracetamol for discomfort.

3. What is the underlying problem, and how will you manage it?

> **Bladder and bowel dysfunction**: constipation and holding cause incomplete emptying. Treat **constipation** (e.g. **lactulose 0.5–1 mL/kg twice daily** or PEG, titrated to soft daily stools), **timed voiding every 2–3 hours** and double voiding, adequate water, feet supported on the toilet, a **letter to the school** about toilet access, and a bladder/bowel diary.

4. Does she need imaging, and which?

> Yes: **recurrent UTI at ≥3 years** (NICE table 6): **ultrasound within 6 weeks** (with pre- and post-void bladder volumes) and **DMSA at 4–6 months**; MCUG not routine.

5. Should she get antibiotic prophylaxis?

> Consider only if UTIs continue despite treating BBD (e.g. trimethoprim 2 mg/kg at night); RIVUR showed prophylaxis reduces recurrences but not scarring and increases resistance. The first-line 'prophylaxis' here is **fixing constipation and voiding habits**.

> **Take-away:** Recurrent cystitis in a school-age child is usually bladder and bowel dysfunction: treat the constipation and voiding habits, image as for recurrent UTI, and reserve prophylaxis.

References

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.